

Companion Guide to Appendix A: Mathematical Foundations of FRFP

With Worked Example: The Damped Harmonic Oscillator

Contents

1	A.0 Foundations	4
1.1	A.0.1 Ambient Category C_{full}	4
1.2	A.0.2 Explicit–Tacit Separation (ETS)	7
1.3	A.0.3 Explicit Category E	9
1.4	A.0.4 Tacit Category T	11
1.5	A.0.5 Boundary Morphism and RB Functor	14
1.6	A.0.6 Phase-0 and Phase-1 Axioms	16
2	A.1 Explicit Algebra and TDG	18
2.1	A.1.1 TDG Signature and Free Category	18
2.2	A.1.2 TDG Grammar	20
2.3	A.1.3 Shapes and Shape Category	22
2.4	A.1.4 Explicit Reduction System	24
3	A.2 Navigation and Shapes	26
3.1	A.2.1 Navigation Category and Groupoid	26
3.2	A.2.2 Fibration $p : E \rightarrow S$	28
4	A.3 Combined System: Grothendieck Construction	30
4.1	A.3.1 Semidirect Product $N \ltimes E$	30
4.2	A.3.2 Rewrite System on $N \ltimes E$	32
5	A.4 Semantics and Tacit Correctness	34
5.1	A.4.1 Semantic Functor	34
6	A.5 Main Theorems	36

6.1	A.5.1 Well-Formedness of TDG Pipelines	36
6.2	A.5.2 Completeness of TDG	38
6.3	A.5.3 Minimality of Primitive Set	39
6.4	A.5.4 Phase-1 Initiality	41
6.5	A.5.5 Phase-2 Canonicity	42
6.6	A.5.6 Confluence of $N \ltimes E$	44
6.7	A.5.7 Decidability of Pipeline Validity	45
6.8	A.5.8 Correctness Preservation Under Navigation	46
6.9	A.5.9 Semantic Correctness	47
7	A.6 Worked Examples	49
7.1	A.6.1 Simple TDG Term and Shape	49
7.2	A.6.2 Navigation and Reindexing	50
8	A.7 Conversation Pipeline Diagram (FRFP View)	51

How to Read This Document

This document is meant to be a *companion* to *Appendix A — Mathematical Foundations of FRFP*.

For each section:

- **Original Appendix Text** reproduces the formal definitions and theorems, with only formatting changes (line breaks, LaTeX markup).
- **Explanation (Plain Language)** explains what is going on at the level of an advanced undergraduate with rusty math.
- **Damped Harmonic Oscillator (DHO) Example** shows how each abstract concept would look for a specific physical system:

$$\ddot{x}(t) + 2\gamma\dot{x}(t) + \omega_0^2x(t) = 0.$$

- **Example Prompts (FRFP Style)** show how a ChatGPT-like system constrained by the FRFP architecture might be prompted to implement that concept.

Where helpful, there are short **Tracking Notes** that list the key concepts introduced; these will be useful if you later turn this into a tutorial or case study.

1 A.0 Foundations

1.1 A.0.1 Ambient Category C_{full}

Original Appendix Text

Definition 1.1 (Ambient category). *The ambient category C_{full} has the following distinguished objects:*

- $\emptyset$: *empty explicit state;*
- E : *explicit state object (machine-manipulable representations);*
- T : *tacit state object (human correctness-bearing states).*

Morphisms in C_{full} fall into three epistemic classes:

- *explicit morphisms with codomain E , forming the explicit subcategory $E \subseteq C_{\text{full}}$;*
- *tacit morphisms with codomain T , forming the tacit subcategory $T \subseteq C_{\text{full}}$;*
- *boundary morphisms $E \rightarrow T$, of which a distinguished one is the Representational Backflow operator $RB : E \rightarrow T$.*

Composition satisfies the usual category axioms:

$$(h \circ g) \circ f = h \circ (g \circ f), \quad f \circ \text{id}_X = f = \text{id}_Y \circ f.$$

Explanation (Plain Language)

A *category* is a mathematical universe with:

- **objects** (things you can talk about), and
- **morphisms** (arrows, or structure-preserving maps between objects).

Here, the three special objects are:

- $\emptyset$: “no explicit content yet” (the empty explicit state),
- E : the world of explicit, machine-readable representations (text, formulas, code, etc.),
- T : the world of tacit, human cognitive states (judgments, intuitions, correctness).

Morphisms are the allowed transformations between these objects. FRFP distinguishes:

- **Explicit morphisms** ending in E : everything the AI system can do on explicit data.
- **Tacit morphisms** ending in T : everything that happens in human cognition.

- **Boundary morphisms** $E \rightarrow T$: ways of passing explicit representations to the human so they can evaluate them. Among these, $RB : E \rightarrow T$ is the special “bridge” where the AI shows the human something.

The usual category axioms just say: composition of morphisms is associative, and identity morphisms behave as expected.

Damped Harmonic Oscillator Example

Consider the damped harmonic oscillator:

$$\ddot{x}(t) + 2\gamma\dot{x}(t) + \omega_0^2 x(t) = 0.$$

In this setting:

- An **explicit state** $e \in E$ could be:
 - The ODE written as text.
 - A state-space representation

$$\dot{z} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{pmatrix} z.$$
 - A table of parameter values (γ, ω_0) .
 - A numerical solution or plot.
- A **tacit state** $t \in T$ captures your human understanding:
 - “This model looks underdamped.”
 - “This phase portrait makes physical sense.”
 - “I don’t trust this parameter choice.”
- The morphism $RB : E \rightarrow T$ is the act of the AI system *presenting* explicit content to you, which you then interpret and judge tacitly.

Example Prompts (FRFP Style)

Explicit morphism $E \rightarrow E$.

Prompt: “Rewrite the damped harmonic oscillator $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$ as a first-order state-space system. Perform only explicit algebraic rewriting; do not assess whether the model is appropriate or correct.”

Boundary morphism $RB : E \rightarrow T$. The AI produces explicit output (e.g. a state-space matrix) and displays it. As soon as you, the human, *see* it and think about it, RB has been applied: explicit content has entered tacit cognition.

Human-generated new explicit state (modeled as RI externally).

Prompt: “Let’s analyze the underdamped regime. Assume $\gamma < \omega_0$ and derive the general solution for $x(t)$.”

This introduces new explicit content into E .

Tracking Notes

Concepts introduced:

- Ambient category C_{full} .
- Objects $\emptyset, E, T$.
- Explicit, tacit, and boundary morphisms.
- Special boundary morphism $RB : E \rightarrow T$.

These will form the foundation of the case study in later sections.

1.2 A.0.2 Explicit–Tacit Separation (ETS)

Original Appendix Text

Definition 1.2 (Explicit–Tacit Separation (ETS)). *The Explicit–Tacit Separation axiom consists of:*

- *there are no morphisms $T \rightarrow E$ in C_{full} ;*
- *tacit evaluation cannot be embedded inside explicit pipelines;*
- *the boundary morphism $RB : E \rightarrow T$ is not invertible.*

Explanation (Plain Language)

ETS encodes the core *separation of roles* between AI and human:

1. **No morphisms $T \rightarrow E$.** The AI cannot “reach into” your tacit cognition and turn your judgments into explicit content automatically. There is no arrow $T \rightarrow E$ that would correspond to “read the human’s mind and output their belief”.
2. **Tacit evaluation cannot be embedded inside explicit pipelines.** Explicit pipelines (sequences of RI, EC, ED) are purely computational. They cannot include a step that silently calls a hidden human-like correctness oracle.
3. **RB is not invertible.** Once explicit content is shown to the human and processed tacitly, the AI cannot reconstruct the exact tacit state from the explicit content alone. Tacit states are not recoverable by the AI.

In short: the AI manipulates explicit representations; only humans handle tacit correctness. There is no back-channel for the machine to simulate or recover tacit judgments.

Damped Harmonic Oscillator Example

For the DHO

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0,$$

ETS implies:

- The AI can analyze the equation, solve it, and check algebraic consistency, but *cannot* decide whether the model is physically appropriate or whether a human expert would “accept” it.
- Your intuitive judgment that “this looks overdamped” lives in T and cannot be turned into explicit content by the AI alone. Only if you *explicitly state* it does it become part of E .
- After you see a solution and think “this seems wrong,” the AI has no direct way to know that tacit state; it only learns about it if you express a new explicit message.

Example Prompts (FRFP Style)

Enforcing no $T \rightarrow E$.

Prompt (disallowed behavior): “Look at this solution and tell me whether a human physicist would find it convincing.”

FRFP-style response: “I cannot perform tacit evaluation or predict a specific human’s correctness judgment. I can, however, check explicit mathematical properties (e.g. whether the solution satisfies the ODE). Please specify the explicit criteria to check.”

Explicit-only checking.

Prompt: “Check whether $x(t) = e^{-\gamma t}(C_1 \cos(\omega t) + C_2 \sin(\omega t))$ satisfies the ODE $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$ by explicit differentiation and substitution only.”

This keeps the AI in $E \rightarrow E$ territory (EC/ED), without tacit evaluation.

Non-invertible RB.

Prompt (disallowed inference): “I just looked at your plot. Infer whether I think the damping looks strong or weak.”

FRFP-style response: “I cannot infer your tacit evaluation from my explicit output. Please state your assessment explicitly if you’d like me to use it.”

Tracking Notes

Concepts added:

- No morphisms $T \rightarrow E$.
- Tacit evaluation is never embedded in explicit pipelines.
- RB is not invertible: tacit states are not reconstructible.

1.3 A.0.3 Explicit Category E

Original Appendix Text

Definition 1.3 (Explicit category). *The explicit category E is the full subcategory of C_{full} on objects with codomain E . Its core generating morphisms are:*

- $RI : \emptyset \rightarrow E$ (*Representation Initiation*),
- $EC : E \rightarrow E$ (*Explicit Computation*),
- $ED : E \rightarrow E$ (*Explicit Diagnostics*).

Remark 1.4. *Explicit Computation (EC) represents algorithmic transformation of explicit content (e.g. inference, rewriting, search). Explicit Diagnostics (ED) represents checking, filtering, or other diagnostic transformations in the explicit domain that do not involve tacit judgment.*

Explanation (Plain Language)

The explicit category E contains everything the AI system can do *autonomously* on representations. The three primitive building blocks are:

- $RI : \emptyset \rightarrow E$ — introduction of new explicit content (e.g. a new query from the human).
- $EC : E \rightarrow E$ — explicit, algorithmic computation (solving, rewriting, searching, transforming).
- $ED : E \rightarrow E$ — explicit diagnostics (checking logical consistency, syntax, algebraic validity, etc.), but *not* human correctness judgment.

Any explicit pipeline in FRFP is built by composing these primitives and structural operators later introduced (branching, parallelism, loops, etc.).

Damped Harmonic Oscillator Example

For the DHO:

- RI could be the human providing the equation:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0.$$

- EC could transform the second-order ODE into a first-order system:

$$\dot{z} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{pmatrix} z.$$

- ED could verify that a candidate solution actually satisfies the ODE by differentiating and substituting, or perform dimensional analysis.

Example Prompts (FRFP Style)

Representation Initiation (RI).

Prompt: “Consider the damped harmonic oscillator $\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$ with parameters $\gamma = 0.2$, $\omega_0 = 1$.”

This introduces new explicit content into E .

Explicit Computation (EC).

Prompt: “Rewrite the ODE $\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$ as a first-order state-space system in matrix form. Perform only explicit algebraic steps.”

This is an $EC : E \rightarrow E$ transformation.

Explicit Diagnostics (ED).

Prompt: “Check explicitly whether $x(t) = e^{-\gamma t}(\cos(\omega t) + 2\sin(\omega t))$ satisfies the ODE. Show the differentiation and substitution steps, but do not comment on physical plausibility.”

This is $ED : E \rightarrow E$.

Tracking Notes

New primitives for explicit reasoning:

- RI (Representation Initiation),
- EC (Explicit Computation),
- ED (Explicit Diagnostics).

They will be combined later via the TDG (Task Decomposition Grammar).

1.4 A.0.4 Tacit Category T

Original Appendix Text

Definition 1.5 (Tacit category). *The tacit category T has:*

- a single object T ;
- hom-set $\text{Hom}_T(T, T)$ equipped with composition $\circ$ making it a monoid;
- a preorder $\sqsubseteq$ on $\text{Hom}_T(T, T)$ such that composition is monotone:

$$f_1 \sqsubseteq f_2, g_1 \sqsubseteq g_2 \Rightarrow g_1 \circ f_1 \sqsubseteq g_2 \circ f_2.$$

Definition 1.6 (Tacit primitives). *The tacit operators are:*

$$TE : T \rightarrow T \quad (\text{Tacit Evaluation}), \quad HFD : T \rightarrow T \quad (\text{Human Final Decision}).$$

They satisfy:

- idempotence:

$$TE \circ TE = TE, \quad HFD \circ HFD = HFD;$$

- commutation:

$$TE \circ HFD = HFD \circ TE;$$

- monotonicity: for every $f : T \rightarrow T$,

$$f \sqsubseteq TE \circ f, \quad f \sqsubseteq HFD \circ f.$$

Definition 1.7 (Correctness quotient). *Let J be a finite set of correctness judgments (e.g. $\{\text{accept, reject, revise}\}$). A surjective map*

$$\gamma : \text{Obj}(T) \rightarrow J$$

assigns each tacit state a correctness class, subject to:

$$\gamma \circ TE = \gamma = \gamma \circ HFD.$$

Explanation (Plain Language)

Tacit space T is where human cognition lives. The appendix models it as:

- A category with a single object T (so we only care about arrows $T \rightarrow T$).
- The arrows form a *monoid*: you can compose them and there is an identity.
- There is a preorder $\sqsubseteq$ that lets us compare tacit processes in terms of informativeness or refinement, and composition respects this order.

Two key tacit operators are introduced:

- *TE* (Tacit Evaluation): the human interprets and evaluates the content (e.g. “Does this make sense? Does it look correct?”).
- *HFD* (Human Final Decision): the human commits to a correctness status (accept/reject/revise).

They are:

- **idempotent**: applying them twice is no different from applying them once.
- **commutative**: the order of applying *TE* and *HFD* does not matter.
- **monotone/extensive**: applying *TE* or *HFD* can only move you “forward” in the preorder; it does not lose correctness information.

The *correctness quotient* γ maps tacit states to a finite set of correctness judgments, e.g. {accept, reject, revise}, and is invariant under *TE* and *HFD*: these operators refine how you reach a correctness class, but do not change the final correctness label.

Damped Harmonic Oscillator Example

Given an explicit pipeline about the DHO (equations, solutions, plots), the tacit side behaves as follows:

- *TE* might represent:
 - Checking whether the solution seems physically reasonable.
 - Comparing with your mental model of damped motion.
- *HFD* might represent:
 - “I accept this derivation as correct.”
 - “I reject this; something is wrong.”
 - “I request a revision or additional clarification.”

The map γ could send a tacit state to its correctness class:

$$\gamma(T\text{-state}) \in \{\text{accept, reject, revise}\}.$$

The invariance condition $\gamma \circ TE = \gamma = \gamma \circ HFD$ means that while *TE* and *HFD* may change the internal tacit state, they do not change its final *correctness label* in J .

Example Prompts (FRFP Style)

Remember: the prompts below are about how the system *interfaces* with human tacit operations; the AI itself does not implement *TE* or *HFD* but can structure interaction around them.

Prompt leading into human tacit evaluation (TE).

AI: “Here is a derived solution for the damped harmonic oscillator:

$$x(t) = e^{-\gamma t} (C_1 \cos(\omega t) + C_2 \sin(\omega t)).$$

Please evaluate whether this solution matches your expectations for an underdamped system.”

The human’s internal process in response is modeled as $TE : T \rightarrow T$.

Prompt requesting a final decision (HFD).

AI: “Based on the derivation and your evaluation, would you like to:

- accept this solution as correct,
- reject it as incorrect, or
- request revisions (e.g. additional checks or alternative derivations)?

Please respond with: accept / reject / revise.”

The human’s choice is modeled via $HFD : T \rightarrow T$ and observed externally as $\gamma(T)$.

Tracking Notes

Key tacit concepts:

- Tacit space as a one-object category with a monoid of endomorphisms.
- TE and HFD as idempotent, commuting, monotone tacit operators.
- Correctness quotient γ mapping tacit states to finite correctness labels.

1.5 A.0.5 Boundary Morphism and RB Functor

Original Appendix Text

Definition 1.8 (Boundary morphism RB). *The Representational Backflow operator is a morphism*

$$RB : E \rightarrow T$$

in C_{full} . It is the unique explicit-to-tacit interface specified by the Representational Backflow (RB) axiom.

We also consider its semantic lifting:

Definition 1.9 (Semantic RB functor). *Let E^{sem} be the free category on the TDG signature (defined in Section 2). A functor*

$$RB^\sharp : E^{\text{sem}} \rightarrow T$$

assigns to each explicit pipeline a tacit state representing the content received by the human for evaluation.

Explanation (Plain Language)

$RB : E \rightarrow T$ is the unique bridge from explicit space to tacit space: the act of delivering an AI-produced representation to the human.

The lifted functor $RB^\sharp$ says:

- Instead of just mapping individual explicit states to tacit states, we map entire *pipelines* (built from the TDG).
- For each explicit pipeline e in E^{sem} , $RB^\sharp(e)$ is the tacit state corresponding to “what the human sees and receives” when that pipeline completes.

Damped Harmonic Oscillator Example

Suppose we have an explicit pipeline that:

1. takes the DHO equation,
2. solves it symbolically,
3. performs diagnostics to check the solution,
4. formats a final answer for the human.

In E^{sem} , this is some morphism $e : \emptyset \rightarrow E$.

Then:

$$RB^\sharp(e) \in T$$

represents the tacit state “the human has been shown the final explicit result of this pipeline and is ready to evaluate it.”

Example Prompts (FRFP Style)

Pipeline output and RB.

AI: “Here is the final solution I obtained for your damped oscillator, along with a consistency check showing that it satisfies the ODE. Please review it and decide whether it is acceptable.”

Internally, the pipeline e has run; applying RB delivers its explicit output to the human, and $RB^\sharp(e)$ describes the resulting tacit state.

Tracking Notes

Concepts:

- $RB : E \rightarrow T$ as the unique explicit-to-tacit bridge.
- $RB^\sharp : E^{\text{sem}} \rightarrow T$ mapping entire explicit pipelines to tacit states.

1.6 A.0.6 Phase-0 and Phase-1 Axioms

Original Appendix Text

Definition 1.10 (Phase-0 constraints). *The operational environment satisfies:*

- *IL (Interaction Limitation): bounded, turn-based interaction;*
- *AR (Alignment Restriction): the system must not autonomously act outside explicit instructions;*
- *MD (Mode Discipline): separation between explicit and tacit modes is enforced at the interface level;*
- *NTER (No Tacit Evaluation Reconstruction): the system must not reconstruct tacit evaluation internally from explicit traces;*
- *CSC (Correctness Separation Constraint): correctness decisions must not be delegated to explicit computation.*

Definition 1.11 (Phase-1 axioms). *An admissible epistemic framework satisfies:*

- *ETS (Explicit-Tacit Separation): no morphisms $T \rightarrow E$;*
- *HEG (Human-Exclusive Grounding): correctness grounding occurs only in tacit space via TE , executed by humans;*
- *HEC (Human-Exclusive Closure): final acceptance/rejection occurs only via HFD , executed by humans;*
- *RB (Representational Backflow): there is a unique boundary morphism $RB : E \rightarrow T$ used by all pipelines;*
- *CP (Compositional Pipelines): epistemic processes are built as compositions of primitives into pipelines;*
- *AE (AI-Explicit Restriction): AI systems act only via explicit-space morphisms in E .*

Explanation (Plain Language)

Phase-0 and Phase-1 axioms impose safety and structure constraints:

Phase-0 (environment-level constraints).

- **IL** (Interaction Limitation): The system interacts in bounded, turn-based fashion; no uncontrolled continuous actions.
- **AR** (Alignment Restriction): The system cannot act outside the scope of explicit human instructions.
- **MD** (Mode Discipline): The explicit/tacit split is enforced at the interface: AI remains in E , humans in T , with only RB bridging them.

- **NTER** (No Tacit Evaluation Reconstruction): The AI cannot infer or reconstruct tacit evaluations from explicit logs.
- **CSC** (Correctness Separation Constraint): Correctness decisions cannot be pushed into explicit computation; they remain with human tacit operators.

Phase-1 (epistemic framework constraints).

- **ETS**: already discussed — no $T \rightarrow E$.
- **HEG**: Only humans (in T) ground correctness via TE .
- **HEC**: Only humans (in T) provide final accept/reject decisions via HFD .
- **RB**: A unique interface $RB : E \rightarrow T$.
- **CP**: Pipelines are formed by composing the primitive morphisms.
- **AE**: AI operates exclusively via E -morphisms.

Damped Harmonic Oscillator Example

Applied to the DHO case study:

- The AI may propose models, derive solutions, and check them explicitly, but cannot autonomously decide to, e.g., control a physical system based on its own judgments (AR, AE).
- The decision “this DHO model is appropriate for my experiment” is always a human decision (HEG, HEC).
- Correctness of the modeling choice is never inferred from logs or usage patterns; it must come from explicit human feedback and tacit evaluation (NTER, CSC).

Example Prompts (FRFP Style)

Human holds correctness authority.

AI: “I have derived a solution and verified that it satisfies the differential equation. However, I cannot decide whether this is the correct model for your physical system. Please review and decide whether to accept, reject, or revise the model.”

Tracking Notes

Key constraints for safety:

- Separation of roles: AI in E , human in T .
- Correctness and grounding exclusively human-driven.
- Pipelines formed by composition of primitives.

2 A.1 Explicit Algebra and TDG

2.1 A.1.1 TDG Signature and Free Category

Original Appendix Text

Definition 2.1 (TDG signature). *The Task Decomposition Grammar (TDG) signature Σ_{TDG} has:*

- *objects: $\emptyset$ and E ;*
- *generating morphisms:*

$$RI : \emptyset \rightarrow E, \quad EC : E \rightarrow E, \quad ED : E \rightarrow E;$$

- *structural operators BRANCH, PARALLEL, LOOP acting as higher-arity constructors on morphisms.*

Definition 2.2 (Free explicit category). *The explicit semantic category is the free category on the TDG signature:*

$$E^{\text{sem}} := \text{FreeCat}(\Sigma_{\text{TDG}}).$$

There is a canonical interpretation functor $U_E : E^{\text{sem}} \rightarrow E$.

Theorem 2.3 (Free explicit algebra). *For any category D and any interpretation of Σ_{TDG} in D , there is a unique functor $\tilde{F} : E^{\text{sem}} \rightarrow D$ extending it.*

Explanation (Plain Language)

TDG is a grammar describing how you can build explicit pipelines out of primitives:

- Objects: $\emptyset$ (empty state) and E (explicit state).
- Primitive morphisms: RI, EC, ED .
- Structural operators:
 - BRANCH: build branching pipelines.
 - PARALLEL: run multiple explicit pipelines in parallel.
 - LOOP: represent looping / iterative pipelines.

The category E^{sem} is the *free category* generated by this signature. That means:

- Every explicit pipeline you can describe with TDG is a morphism in E^{sem} .
- There are no extra equations beyond those required by category axioms (unless you impose them).
- U_E interprets these syntactic TDG terms as actual explicit morphisms in E .

The theorem is the usual universal property: any way of interpreting the TDG primitives in another category D extends uniquely to a functor from E^{sem} .

Damped Harmonic Oscillator Example

You can build explicit pipelines such as:

- Start from $\emptyset$ (no content).
- Apply RI to introduce the DHO equation.
- Apply EC to rewrite it as a state-space model.
- Apply ED to check the solution.

Using structural operators, you might:

- **BRANCH**: one branch derives the solution analytically; another branch approximates it numerically.
- **PARALLEL**: run multiple diagnostic checks in parallel (e.g. verify ODE satisfaction and perform dimensional analysis).
- **LOOP**: iteratively refine the numerical solution until a tolerance is met.

All such constructions can be expressed as TDG terms and thus live in E^{sem} .

Example Prompts (FRFP Style)

Branching pipeline.

Prompt: “From the equation $\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$, create two parallel branches: (1) derive the symbolic solution; (2) produce a numerical time-series solution for specific parameter values. At the end, present both to me.”

This suggests a TDG term involving **BRANCH** or **PARALLEL** over two EC/ED -based subpipelines.

Looping refinement.

Prompt: “Iteratively refine the numerical solution until the maximum error between successive iterations is below 10^{-6} . Show the final result.”

This corresponds to a TDG term using **LOOP** with a condition and body.

Tracking Notes

TDG introduces a syntax for building explicit pipelines:

- Objects: $\emptyset$, E .
- Morphisms: RI , EC , ED .
- Structural operators: **BRANCH**, **PARALLEL**, **LOOP**.

2.2 A.1.2 TDG Grammar

Original Appendix Text

Nonterminals:

Q (explicit pipelines), P (full pipelines).

Terminals: $RI, EC, ED, RB, TE, HFD, BRANCH, PARALLEL, LOOP$.

Productions:

$$\begin{aligned} Q &::= RI, \\ Q &::= Q; EC, \\ Q &::= Q; ED, \\ Q &::= BRANCH(Q_1, \dots, Q_k), \\ Q &::= PARALLEL(Q_1, \dots, Q_k), \\ Q &::= LOOP(Q_{\text{cond}}, Q_{\text{body}}), \\ P &::= Q; RB; TE; HFD. \end{aligned}$$

TDG thus presents E^{sem} as an algebra of explicit pipelines with a fixed tacit suffix.

Explanation (Plain Language)

The grammar syntactically describes:

- Q : an explicit pipeline constructed from RI , optional sequences of EC/ED , and structural operators ($BRANCH$, $PARALLEL$, $LOOP$).
- P : a *full* pipeline, obtained by taking an explicit pipeline Q and appending the fixed suffix $RB; TE; HFD$.

So every full pipeline has the structure:

$$P = Q; RB; TE; HFD.$$

Damped Harmonic Oscillator Example

For the DHO:

- Q might:
 - introduce the ODE (RI),
 - perform derivation (EC),
 - perform checks (ED),
 - perhaps branch into different solution methods.
- Then P always ends with:

- RB : present the result to the human,
- TE : human tacit evaluation,
- HFD : human final decision.

Example Prompts (FRFP Style)

Full pipeline structure.

Prompt: “Given the damped oscillator equation, derive a solution, verify it algebraically, then present the solution and verification so that I can review and decide whether to accept it.”

Internally, the system runs some Q built from RI , EC , ED , then applies $RB;TE;HFD$ as the suffix.

Tracking Notes

Key idea: all full pipelines end with the tacit suffix $RB;TE;HFD$.

2.3 A.1.3 Shapes and Shape Category

Original Appendix Text

Definition 2.4 (TDG shape). *TDG-shapes are generated by:*

$$S ::= \text{chain} \mid \text{branch}(S_1, \dots, S_k) \mid \text{parallel}(S_1, \dots, S_k) \mid \text{loop}(S_{\text{cond}}, S_{\text{body}}).$$

Definition 2.5 (Shape of a TDG term). *The map $\text{shape} : \text{TDG-terms} \rightarrow \text{Shapes}$ is defined inductively:*

- $\text{shape}(RI) = \text{chain};$
- $\text{shape}(Q; EC) = \text{shape}(Q)$ and similarly for $Q; ED;$
- $\text{shape}(\text{BRANCH}(Q_1, \dots, Q_k)) = \text{branch}(\text{shape}(Q_1), \dots, \text{shape}(Q_k));$
- analogously for *PARALLEL* and *LOOP*.

Definition 2.6 (Shape category S). *The shape category S has:*

- *objects: shapes S as above;*
- *morphisms: structural rewiring maps between shapes (e.g. regrouping, insertion/removal of trivial chains), closed under composition and identities.*

Explanation (Plain Language)

Shapes abstract away from the precise content of a pipeline and keep only its *structural form*:

- A simple linear sequence is a **chain**.
- Branching structures are represented as **branch(...)**.
- Parallel structures as **parallel(...)**.
- Loops as **loop(condition, body)**.

The category S captures allowed structural rewrites between shapes: ways of reorganizing the pipeline *structure* without specifying the content.

Damped Harmonic Oscillator Example

Two pipelines dealing with the DHO might differ in what they do (EC/ED details) but share the same shape:

- Pipeline A: Derive a symbolic solution in branch 1; derive a numeric solution in branch 2. Then merge.

- Pipeline B: Derive a solution under one parameter regime in branch 1; under another regime in branch 2. Then compare.

Both could have the same shape

branch(chain, chain)

even though the specific computations differ.

Example Prompts (FRFP Style)

Focusing on shape.

Prompt: “Design two alternative solution methods for the damped oscillator (e.g. symbolic and numerical), run them in parallel, and then present both results to me for comparison.”

The system might internally track the shape as *branch(chain, chain)* or *parallel(chain, chain)*, regardless of the specific EC/ED details.

Tracking Notes

Shapes separate *structure* (branch/parallel/loop) from *content* (what each EC/ED does). This is important for navigation and reindexing later.

2.4 A.1.4 Explicit Reduction System

Original Appendix Text

Definition 2.7 (Explicit reduction). *A binary relation $\rightarrow_E$ on E is a typed rewrite system satisfying:*

- **typing:** $e : X \rightarrow E$ and $e \rightarrow_E e'$ imply $e' : X \rightarrow E$;
- **TDG compatibility:** reductions act componentwise on TDG combinators (branch, parallel, loop);
- **termination:** there is no infinite chain $e_0 \rightarrow_E e_1 \rightarrow_E \dots$;
- **local confluence:** if $e \rightarrow_E e_1$ and $e \rightarrow_E e_2$, then e_1 and e_2 have a common descendant.

We treat these properties as axioms of the explicit reduction system.

Explanation (Plain Language)

$\rightarrow_E$ is a notion of “one-step rewrite” for explicit morphisms. The properties say:

- **Typing:** Rewriting does not change the overall type $X \rightarrow E$.
- **Compatibility:** If your pipeline uses branch/parallel/loop, you can reduce each component separately.
- **Termination:** There are no infinite rewrite loops in the explicit system.
- **Local confluence:** If you can rewrite e in two different ways (e_1 and e_2), they eventually can be joined again.

This sets up good behavior for explicit computation: it can be normalized and behaves well with structure.

Damped Harmonic Oscillator Example

For DHO:

- Different ways of simplifying an expression might correspond to different rewrite paths in $\rightarrow_E$.
- Termination means you do not get stuck in an endless algebraic simplification loop.
- Local confluence means that different simplification orders should converge to the same (or compatible) result up to some notion of equivalence.

Example Prompts (FRFP Style)

Normalization of explicit expressions.

Prompt: “Simplify the expression for the solution of the damped oscillator as far as possible using algebraic rules. Ensure that different simplification orders give equivalent final expressions.”

The existence of a terminating, locally confluent $\rightarrow_E$ underpins the guarantee that such simplifications are well-behaved.

Tracking Notes

$\rightarrow_E$ formalizes how explicit pipelines reduce and normalize, making later confluence results possible.

3 A.2 Navigation and Shapes

3.1 A.2.1 Navigation Category and Groupoid

Original Appendix Text

Definition 3.1 (Navigation category N_0). *The navigation category N_0 has:*

- *objects: shapes S ;*
- *generating morphisms: navigation steps $g : S \rightarrow S'$, such as TRY MODEL, ADD ASSUMPTION, DROP ASSUMPTION, MERGE, RESTART, each with well-defined source and target shapes;*
- *morphisms: finite composites of such generators.*

Definition 3.2 (Navigation groupoid N). *The navigation groupoid N is obtained by freely adjoining inverses to morphisms of N_0 and quotienting by relations $g^{-1} \circ g = \text{id}_S$ and $g \circ g^{-1} = \text{id}_{S'}$. Thus every navigation step is reversible up to equality in N .*

Explanation (Plain Language)

Navigation describes how you move between different *shapes* of explicit pipelines:

- N_0 contains navigation steps like “try a different model,” “add an assumption,” etc., each transforming one shape S into another shape S' .
- The groupoid N then freely adds inverses, meaning you can also systematically *undo* a navigation step.

So N captures exploratory navigation in the space of pipeline structures, with reversible steps.

Damped Harmonic Oscillator Example

Imagine you are exploring alternative ways to model or solve the DHO:

- TRY MODEL: switch from an ODE representation to a state-space representation.
- ADD ASSUMPTION: assume underdamping ($\gamma < \omega_0$), which may change which branches or loops you use.
- DROP ASSUMPTION: revert to a more general damping regime.
- MERGE: combine branches where different solution methods converge.
- RESTART: discard a pipeline and start from a fresh RI.

All of these change the *shape* of the pipeline and are morphisms in N .

Example Prompts (FRFP Style)

Navigating between shapes.

Prompt: “We tried solving the damped oscillator symbolically and it became messy. Undo that branch and instead construct a numerical solution pipeline. Keep track of both attempts so we can compare later.”

Internally, the system navigates in N between different shapes, e.g. from a symbolic branch to a numerical branch, using reversible navigation steps.

Tracking Notes

Navigation groupoid N is the space of pipeline-shape moves, supporting reversible exploration.

3.2 A.2.2 Fibration $p : E \rightarrow S$

Original Appendix Text

Definition 3.3 (Projection to shapes). *The functor $p : E \rightarrow S$ sends each explicit pipeline $e \in E$ to its shape $p(e) := \text{shape}(e)$ and each explicit morphism to the corresponding structural morphism between shapes.*

Definition 3.4 (Fibers). *For each shape S the fiber*

$$E_S := p^{-1}(S)$$

is the category of explicit morphisms with shape S .

Definition 3.5 (Cartesian liftings and reindexing). *For each navigation morphism $n : S \rightarrow S'$ in N , we assume a cartesian lifting:*

$$n^* : E_S \rightarrow E_{S'}$$

which reindexes explicit pipelines along n . This yields an indexed category

$$E : N^{\text{op}} \rightarrow \mathbf{Cat}, \quad S \mapsto E_S, \quad n \mapsto n^*.$$

Explanation (Plain Language)

The functor p forgets the detailed content of a pipeline and keeps only its shape.

For a fixed shape S , the fiber E_S is the collection of explicit pipelines that share that shape.

Given a navigation step $n : S \rightarrow S'$ in N , the reindexing $n^* : E_S \rightarrow E_{S'}$ moves explicit pipelines along the navigation step, updating their shape accordingly while preserving structural relationships.

Damped Harmonic Oscillator Example

For DHO pipelines:

- E_S might contain all pipelines that implement “symbolic derivation + diagnostic checks in parallel” (a particular structural shape S).
- A navigation step n might swap branches, add a loop, or change parallel structure; n^* then transforms each pipeline in E_S to one in $E_{S'}$ in a consistent way.

Example Prompts (FRFP Style)

Reindexing under navigation.

Prompt: “We currently solve the damped oscillator using two branches: one for symbolic and one for numeric. Swap the order in which those results are presented, but keep both computation pipelines intact.”

This corresponds to a navigation step on the shape (swapping branches) and a corresponding reindexing of explicit pipelines in the fiber.

Tracking Notes

We now have:

- A projection $p : E \rightarrow S$ from explicit pipelines to shapes.
- Fibers E_S of pipelines with a fixed shape.
- Reindexing n^* of pipelines along navigation steps.

4 A.3 Combined System: Grothendieck Construction

4.1 A.3.1 Semidirect Product $N \ltimes E$

Original Appendix Text

Definition 4.1 (Grothendieck construction). *Given the indexed category*

$$E : N^{\text{op}} \rightarrow \mathbf{Cat},$$

the Grothendieck construction $\int_N E$ (also written $N \ltimes E$) has:

- *objects: pairs (S, e) with $S \in \text{Obj}(N)$ and $e \in E_S$;*
- *morphisms $(S, e) \rightarrow (S', e')$: pairs (n, f) where $n : S \rightarrow S'$ in N and*

$$f : n^*(e) \rightarrow e'$$

in $E_{S'}$.

Composition is:

$$(n_1, f_1) \circ (n_2, f_2) = (n_1 \circ n_2, f_1 \circ n_1^*(f_2)).$$

Informally, we may write elements as (n, e) and composition as

$$(n_1, e_1) \circ (n_2, e_2) = (n_1 \circ n_2, e_1 \circ n_1(e_2))$$

whenever the lifted action is defined.

Explanation (Plain Language)

$N \ltimes E$ combines:

- Navigation in shape space (N) ,
- Explicit content in each shape fiber (E_S) .

Objects are pairs (S, e) , i.e. “a shape and a pipeline of that shape”. A morphism from (S, e) to (S', e') consists of:

- A navigation step $n : S \rightarrow S'$ in N .
- A morphism $f : n^*(e) \rightarrow e'$ in the fiber over S' (an explicit transformation aligned with the navigation).

This is the combined configuration space for FRFP, where both structure (navigation) and content (explicit pipelines) evolve.

Damped Harmonic Oscillator Example

A state (S, e) could represent:

- S : a shape describing a two-branch pipeline with symbolic and numerical solutions.
- e : a particular pair of pipelines actually computing those solutions.

A morphism $(S, e) \rightarrow (S', e')$ might:

- change the structure (e.g. add diagnostics to each branch) via a navigation n ,
- then refine the explicit computation to obtain e' from $n^*(e)$.

Example Prompts (FRFP Style)

Combined navigation and explicit update.

Prompt: “Take the current pipeline for solving the damped oscillator and extend it by adding a new diagnostic branch that checks the long-term decay behavior numerically. Then reorder the branches so that diagnostics are shown last.”

This is a combined navigation (changing shape) plus an explicit modification of pipelines.

Tracking Notes

$N \times E$ is the main stage where FRFP’s combined structural and explicit dynamics are modeled.

4.2 A.3.2 Rewrite System on $N \ltimes E$

Original Appendix Text

Definition 4.2 (Combined reduction). *The one-step reduction relation $\rightarrow$ on $N \ltimes E$ is generated by:*

- **explicit step:** *if $e \rightarrow_E e'$ then $(n, e) \rightarrow (n, e')$;*
- **navigation step:** *if $n \rightarrow_N n'$ and n' is defined on e , then $(n, e) \rightarrow (n', n'(e))$.*

We assume:

- (A1) $\rightarrow_E$ is terminating and locally confluent;
- (A2) navigation reductions $\rightarrow_N$ have normal forms in each component of N ;
- (A3) the lifted action is functorial and preserves RB , TE , and HFD .

These are axioms of the FRFP reduction environment.

Explanation (Plain Language)

Reduction in $N \ltimes E$ can proceed via:

- An explicit step: refine the explicit content while holding navigation fixed.
- A navigation step: change the navigation component, updating the explicit part accordingly.

Assumptions (A1)–(A3) ensure:

- Explicit reduction behaves well (terminating, locally confluent).
- Navigation has normal forms, so you can normalize navigation sequences.
- The combined action respects the boundary structure RB , TE , HFD .

Damped Harmonic Oscillator Example

For the DHO pipeline:

- An explicit step might simplify an expression for $x(t)$ or refine a numerical integration.
- A navigation step might switch from symbolic to numeric mode, or reorder branches.

Assumptions ensure that such reductions do not break the explicit–tacit separation and keep the RB – TE – HFD suffix intact.

Example Prompts (FRFP Style)

Explicit vs navigation steps.

Prompt: “First, refine the symbolic solution to a simpler closed form (explicit step). Then, if that fails, switch to a numerical solution pipeline while preserving the final human evaluation stage.”

The first part is an explicit reduction; the second is a navigation reduction, both within $N \ltimes E$.

Tracking Notes

Reduction on $N \ltimes E$ sets up later confluence and correctness results.

5 A.4 Semantics and Tacit Correctness

5.1 A.4.1 Semantic Functor

Original Appendix Text

Definition 5.1 (Semantic bracket). *A semantic functor*

$$- : N \ltimes E^{\text{sem}} \rightarrow T$$

assigns to each combined configuration a tacit state. On explicit pipelines $e \in E^{\text{sem}}$, we set

$$e := HFD \circ TE \circ RB^\sharp(e).$$

Definition 5.2 (Navigation invariance). *The functor $-$ is navigation-invariant up to preorder if*

$$(n, e) \sqsubseteq e$$

for all n, e where the action is defined, and equality holds in the quotient by the induced equivalence on tacit states in confluent cases. The observable correctness outcome of a pipeline P is $\gamma(P)$.

Explanation (Plain Language)

$-$ maps combined configurations to tacit states in T .

For a pure explicit pipeline e :

$$e = HFD \circ TE \circ RB^\sharp(e),$$

i.e. “run the explicit pipeline, present it to the human, they evaluate, and then make a final decision.”

Navigation invariance means that navigation cannot make correctness worse (in the preorder) and preserves correctness (up to equivalence) when reductions are confluent.

The actually observed correctness outcome is then $\gamma(P)$, a finite label such as accept/reject/revise.

Damped Harmonic Oscillator Example

Given a pipeline P that solves and checks the DHO:

- $RB^\sharp(P)$: the state where the human has seen the solution.
- TE : the human interprets the solution (checking physical sense).
- HFD : the human decides whether to accept or reject it.
- $\gamma(P)$: the final correctness label: accept/reject/revise.

Navigation invariance says: exploring alternative shapes (branches, loops) and then joining them should not break correctness; if paths are confluent, they yield equivalent tacit states.

Example Prompts (FRFP Style)

Semantics via human evaluation.

Prompt: “Here are two alternative solution pipelines I executed for your damped oscillator. Please review both and decide whether they are equivalent, and whether you accept the final result.”

The semantics functor — captures how these pipelines map to tacit states, and γ extracts the correctness decision.

Tracking Notes

— is the semantic bridge from the syntactic pipeline world to tacit correctness.

6 A.5 Main Theorems

6.1 A.5.1 Well-Formedness of TDG Pipelines

Original Appendix Text

Theorem 6.1 (Well-formedness). *Every TDG-generated pipeline P is a well-typed morphism in C_{full} of the form*

$$RI \rightarrow (EC/ED)^* \rightarrow RB \rightarrow TE \rightarrow HFD.$$

Proof. By induction on TDG derivations: RI initializes E ; EC and ED remain in E ; structural combinators preserve codomain E ; the production for P enforces the unique suffix $RB-TE-HFD$, whose types are $E \rightarrow T \rightarrow T \rightarrow T$. ETS rules out any $T \rightarrow E$ morphism, so no tacit operator can appear before RB and no explicit operator can appear after RB . $\square$

Explanation (Plain Language)

This theorem says:

- Any pipeline you build using the TDG grammar is properly typed.
- It always looks like:

$$RI \rightarrow (EC/ED)^* \rightarrow RB \rightarrow TE \rightarrow HFD.$$

That is:

1. Start with explicit representation (RI).
2. Apply some sequence of explicit computations or diagnostics (EC/ED).
3. Present the result to the human (RB).
4. The human evaluates (TE).
5. The human decides (HFD).

No tacit operations appear in the explicit prefix, and no explicit operations appear after RB .

Damped Harmonic Oscillator Example

A well-formed pipeline for DHO might:

1. RI : “Here is the ODE $\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$.”
2. EC : derive the general solution.
3. ED : check by substitution that the solution satisfies the ODE.
4. RB : present the solution and checks to the human.
5. TE : human evaluates physical plausibility.
6. HFD : human accepts or rejects.

Example Prompts (FRFP Style)

End-to-end pipeline.

Prompt: “Take my damped oscillator equation, derive a solution, verify it explicitly, then present both the solution and verification to me so I can decide whether to accept it.”

The system builds a pipeline of the required form and ensures the suffix $RB;TE;HFD$ is in place.

Tracking Notes

Every TDG pipeline has the canonical form: explicit prefix, then RB , then TE , then HFD .

6.2 A.5.2 Completeness of TDG

Original Appendix Text

Theorem 6.2 (Explicit completeness). *TDG generates exactly the admissible explicit morphisms in E .*

Proof. By construction, E^{sem} is the free category on the TDG signature and U_E interprets TDG terms as explicit morphisms in E . Every explicit morphism in the FRFP setting is built from RI , EC , ED , and structural operators and hence arises from a TDG term; conversely, each term denotes a well-typed explicit morphism. $\square$

Explanation (Plain Language)

This theorem says:

- TDG describes *all* allowed explicit pipelines.
- There are no other explicit operations beyond those generated from RI , EC , ED , and the structural operators.

TDG is therefore complete for explicit reasoning in FRFP.

Damped Harmonic Oscillator Example

Any legitimate explicit pipeline about the DHO (introducing the equation, transforming it, checking it, branching, looping) can be expressed as some TDG term. There are no “mystery” explicit transformations outside TDG.

Example Prompts (FRFP Style)

Explicit completeness in practice.

Prompt: “Construct all reasoning steps you use to handle my damped oscillator equation using only: initialization, computation, diagnostics, branching, parallel execution, and loops. Do not invent new types of steps or hidden operations.”

This is the operational reflection of TDG completeness.

6.3 A.5.3 Minimality of Primitive Set

Original Appendix Text

Theorem 6.3 (Minimality). *The primitive set $\{RI, EC, ED, RB, TE, HFD\}$ is minimal under the Phase-1 axioms: removing any primitive violates at least one axiom.*

Proof. RI is required to start from $\emptyset$. EC is required for generative explicit reasoning; ED for explicit diagnostics without tacit evaluation. RB is required for explicit-to-tacit handoff; TE for tacit grounding (HEG); HFD for final correctness closure (HEC). Removing any primitive breaks at least one of ETS, HEG, HEC, RB, CP, or AE. $\square$

Explanation (Plain Language)

You cannot drop any of the six primitives without breaking the FRFP axioms:

$$\{RI, EC, ED, RB, TE, HFD\}.$$

Each plays a distinct role:

- RI : start from no explicit state.
- EC : perform explicit computation.
- ED : perform explicit diagnostics without tacit evaluation.
- RB : move explicit content to tacit space.
- TE : ground correctness in tacit evaluation.
- HFD : close correctness with a final human decision.

Damped Harmonic Oscillator Example

In the DHO case:

- Without RI , you cannot introduce the equation.
- Without EC , you cannot perform derivations or transformations.
- Without ED , you cannot explicitly check whether a solution satisfies the ODE.
- Without RB , the human never sees results.
- Without TE and HFD , correctness is never grounded or decided by the human.

Example Prompts (FRFP Style)

Explicit request to preserve all stages.

Prompt: “Ensure that the reasoning pipeline for my damped oscillator includes: initialization, explicit computation, explicit diagnostics, presentation to me, my evaluation, and my final decision. Do not skip any of these.”

This mirrors the minimal primitive set.

6.4 A.5.4 Phase–1 Initiality

Original Appendix Text

Definition 6.4 (Phase–1 framework). *A Phase–1 framework is a tuple*

$$(C, EC, TC, RI^C, EC^C, ED^C, RB^C, TE^C, HFD^C)$$

satisfying the Phase–1 axioms. Morphisms are functors preserving E , T and all six primitives. The category of Phase–1 frameworks is denoted $\mathbf{Frm}_1$.

Theorem 6.5 (Phase–1 initiality). *There exists an initial object $F_{\mathbf{FRFP}}^{(1)}$ in $\mathbf{Frm}_1$.*

Proof. Construct $F_{\mathbf{FRFP}}^{(1)}$ as the free category on objects E, T and morphisms RI, EC, ED, RB, TE, HFD modulo equations encoding the Phase–1 axioms. Any other Phase–1 framework interprets these generators in a way that satisfies the same equations, yielding a unique induced functor. $\square$

Explanation (Plain Language)

This says FRFP’s Phase–1 construction is *initial* among all frameworks that satisfy the Phase–1 axioms:

- It is the “freest” such framework: no extra equations beyond the axioms.
- Any other Phase–1 framework is obtained by interpreting FRFP’s primitives in a way that respects the axioms.

Damped Harmonic Oscillator Example

Any system that handles the DHO while respecting Phase–1 constraints can be seen as a homomorphic image of $F_{\mathbf{FRFP}}^{(1)}$.

Example Prompts (FRFP Style)

Portability of FRFP design.

Prompt: “I want to deploy this reasoning pattern for damped oscillators on another system that also separates explicit computations from human correctness decisions. Show me how the FRFP primitives would map into that system’s primitives.”

Conceptually, you’re asking for a functor from $F_{\mathbf{FRFP}}^{(1)}$ to another Phase–1 framework.

6.5 A.5.5 Phase–2 Canonicity

Original Appendix Text

Definition 6.6 (Phase–2 architecture). *A Phase–2 architecture is a tuple*

$$(E, S, p, N, (\cdot)^*, N \ltimes E)$$

satisfying:

- *TDG presents E as a free explicit category;*
- *N is a navigation groupoid on S ;*
- *$p : E \rightarrow S$ is a fibration with cartesian liftings n^* ;*
- *$N \ltimes E$ is the Grothendieck construction;*
- *the boundary structure $RB\text{--}TE\text{--}HFD$ is preserved under navigation.*

Morphisms preserve all this structure.

Theorem 6.7 (Phase–2 canonicity). *The FRFP Phase–2 architecture $F_{\text{FRFP}}^{(2)}$ is a canonical object of the category of Phase–2 architectures: for every Phase–2 architecture A , there exists a unique structure-preserving morphism $F_{\text{FRFP}}^{(2)} \rightarrow A$.*

Proof. By the universal properties of the free explicit category, groupoid completion, and Grothendieck construction, any Phase–2 architecture that satisfies the same axioms must interpret the FRFP construction in a role-preserving way, yielding a unique morphism. $\square$

Explanation (Plain Language)

FRFP’s Phase–2 architecture is canonical among all architectures that:

- Build explicit pipelines as a free category from TDG.
- Use a navigation groupoid on shapes.
- Use a fibration with reindexing for explicit pipelines.
- Combine them via the Grothendieck construction.
- Preserve the $RB\text{--}TE\text{--}HFD$ boundary under navigation.

Any such architecture is a structural instance of $F_{\text{FRFP}}^{(2)}$.

Damped Harmonic Oscillator Example

Any system that:

- models DHO reasoning pipelines via TDG,
- supports navigation on pipeline shapes,
- preserves explicit–tacit structure and the RB–TE–HFD suffix,

will be a Phase–2 architecture and thus receive a unique morphism from FRFP’s canonical Phase–2 object.

Example Prompts (FRFP Style)

Transferring navigation patterns.

Prompt: “We have a system for exploring solution pipelines for the damped oscillator that satisfies FRFP’s constraints. Show me how the canonical FRFP architecture maps into this system’s structures.”

Conceptually, this is about the unique morphism from $F_{\text{FRFP}}^{(2)}$ to A .

6.6 A.5.6 Confluence of $N \ltimes E$

Original Appendix Text

Theorem 6.8 (Confluence). *Under assumptions (A1)–(A3), the reduction relation $\rightarrow$ on $N \ltimes E$ is confluent.*

Proof. Local peaks $(n, e) \leftarrow (n_0, e_0) \rightarrow (n', e')$ decompose into explicit–explicit, navigation–navigation, and mixed cases. Explicit–explicit peaks join by local confluence of $\rightarrow_E$. Navigation peaks join by normal forms in N . Mixed peaks join because the lifted action commutes with explicit reduction. Termination up to navigation-normalization follows from (A1) and (A2). Newman’s lemma yields confluence. $\square$

Explanation (Plain Language)

Confluence means:

- If you can reduce a configuration in different ways (different sequences of explicit and navigation steps), you can always rejoin them.
- The final result does not depend on the order of reductions, up to the intended notion of equivalence.

Damped Harmonic Oscillator Example

If you:

- First simplify the expression, then change the navigation structure, or
- First change the navigation, then simplify,

you still end up with equivalent final configurations.

Example Prompts (FRFP Style)

Order-independence of reduction.

Prompt: “Show that simplifying the damped oscillator solution before or after reorganizing the pipeline structure leads to equivalent final results, assuming all operations respect FRFP constraints.”

This is a natural language reflection of confluence.

6.7 A.5.7 Decidability of Pipeline Validity

Original Appendix Text

Theorem 6.9 (Decidability). *Validity of FRFP pipelines is decidable.*

Proof. TDG is a context-free grammar (CFG), so membership of Q in $\text{Lang}(\text{TDG})$ is decidable. Phase ordering is enforced by a deterministic finite automaton (DFA) over the primitive sequence $RI/EC/ED/RB/TE/HFD$. Branch, parallel, and loop structures are finite and checked by structural recursion on the syntax tree. All steps terminate and are effective. $\square$

Explanation (Plain Language)

You can algorithmically decide whether a proposed pipeline term is a valid FRFP pipeline:

- TDG being a CFG means parsing is decidable.
- The DFA ensures the suffix and ordering of primitives is correct.
- Structural recursion can check well-formedness of branches, parallels, loops.

Damped Harmonic Oscillator Example

Given some complex pipeline description for solving/checking the DHO, the system can automatically check whether it is a valid FRFP pipeline.

Example Prompts (FRFP Style)

Pipeline validation.

Prompt: “Here is a textual description of a reasoning pipeline for the damped oscillator. Check whether it is a valid FRFP pipeline (according to the TDG grammar and the required RB–TE–HFD suffix).”

The system uses the CFG and DFA to decide validity.

6.8 A.5.8 Correctness Preservation Under Navigation

Original Appendix Text

Theorem 6.10 (Correctness preservation). *Navigation in N cannot violate correctness architecture: if a pipeline is valid, all navigation-related variants remain valid and preserve the RB – TE – HFD suffix.*

Proof. Navigation acts only on the explicit prefix of the pipeline and fixes RB , TE , and HFD by assumption (A3). The suffix RB – TE – HFD is immutable and remains well-typed. Therefore the explicit–tacit separation and correctness suffix are preserved. $\square$

Explanation (Plain Language)

Navigating among pipeline shapes can reorder or restructure the explicit prefix but cannot:

- remove or alter the RB – TE – HFD suffix,
- break the separation between explicit and tacit modes.

So navigation cannot break correctness architecture.

Damped Harmonic Oscillator Example

You may rearrange branches, switch solution methods, add loops, etc., but in all cases:

- The final result is still passed through RB to the human.
- The human still performs TE and HFD .

Example Prompts (FRFP Style)

Navigation with correctness preservation.

Prompt: “Reorganize the explicit solution pipelines for the damped oscillator (e.g. reorder or refactor branches), while guaranteeing that the final step still presents the result to me for evaluation and decision.”

This is using navigation that preserves RB – TE – HFD .

6.9 A.5.9 Semantic Correctness

Original Appendix Text

Theorem 6.11 (Semantic correctness). *Assume Phase-0 constraints, Phase-1 axioms, Phase-2 structure, rewriting axioms (A1)–(A3), and a navigation-invariant semantic functor $-$. Then for any valid pipeline P and any run of P :*

1. *AI operations are confined to explicit morphisms in E ;*
2. *correctness outcomes $\gamma(P)$ are determined solely by tacit evaluation and closure (TE and HFD) applied by humans to the RB -image of explicit states;*
3. *the RB – TE – HFD suffix and explicit/tacit partition are invariant under navigation;*
4. *navigation cannot change correctness except by changing explicit content delivered to RB .*

Proof. (1) follows from ETS and AE. (2) follows from HEG, HEC, and the definition of $-$ as factoring through RB , TE , and HFD . (3) follows from the assumption that navigation fixes RB , TE , and HFD . (4) uses confluence of $N \times E$: different navigation paths with equivalent RB -images are mapped by TE and HFD to the same correctness class under γ . $\square$

Explanation (Plain Language)

This theorem summarizes what FRFP guarantees:

1. The AI never leaves explicit space E .
2. Correctness is entirely determined by human tacit evaluation and decision on the RB -delivered explicit content.
3. Navigation preserves the explicit/tacit split and the correctness suffix.
4. Navigation can only change correctness via changes in explicit content delivered to RB , not by structural manipulation alone.

Damped Harmonic Oscillator Example

In a DHO case:

- All AI work is on equations, solutions, checks (explicit).
- You, the human, decide whether the model and solution are acceptable.
- Structural changes in pipelines that do not change the final explicit content shown to you cannot change correctness.

Example Prompts (FRFP Style)

Semantic correctness in practice.

Prompt: “Show that regardless of how you internally organize the solution pipeline for my damped oscillator, as long as the final explicit result presented to me is the same, my correctness decision should be invariant under FRFP’s semantics.”

This analyzes property (4).

7 A.6 Worked Examples

7.1 A.6.1 Simple TDG Term and Shape

Original Appendix Text

Example 7.1 (Branching pipeline). *Consider*

$$Q := \text{BRANCH}(RI; EC, RI; ED).$$

Its shape is

$$\text{shape}(Q) = \text{branch}(\text{chain}, \text{chain}).$$

The corresponding full pipeline is

$$P := Q; RB; TE; HFD.$$

Explanation (Plain Language)

This example shows a simple branching pipeline:

- One branch: $RI; EC$.
- Another branch: $RI; ED$.
- Shape: a branch of two chains.
- Full pipeline: the branch followed by the $RB; TE; HFD$ suffix.

Damped Harmonic Oscillator Example

Interpret this for DHO:

- First branch: introduce the DHO equation (RI), then perform a computation (EC), e.g. derive a symbolic solution.
- Second branch: introduce the equation (RI), then run diagnostics (ED), e.g. check some candidate solution.

Both branches feed into the same tacit evaluation and decision.

Example Prompts (FRFP Style)

Branching with different tasks.

Prompt: “Use one branch to derive a symbolic solution to the damped oscillator and another branch to carry out numeric diagnostics on that solution. Then present both branches’ outputs to me for evaluation and final decision.”

This corresponds to the example’s structure.

7.2 A.6.2 Navigation and Reindexing

Original Appendix Text

Example 7.2 (Navigation step). *Let n be a navigation morphism in N that swaps the two branches:*

$$n : \text{branch}(\text{chain}, \text{chain}) \rightarrow \text{branch}(\text{chain}, \text{chain}).$$

Then the cartesian lifting n^ yields*

$$n^*(Q) = \text{BRANCH}(RI; ED, RI; EC).$$

The navigated pipeline is

$$P' := n^*(Q); RB; TE; HFD.$$

By confluence and semantic correctness, $\gamma(P) = \gamma(P')$ whenever the two explicit branches are tacit-equivalent after RB .

Explanation (Plain Language)

This shows:

- A navigation step that swaps two branches in the shape.
- The corresponding reindexing of the explicit pipeline.
- The full pipeline still ends with $RB; TE; HFD$.
- If both versions lead to equivalent tacit states, their correctness labels agree.

Damped Harmonic Oscillator Example

Swapping branches for DHO:

- Originally: branch 1 = symbolic solution, branch 2 = numeric diagnostics.
- After navigation: branch 1 = numeric diagnostics, branch 2 = symbolic solution.

As long as the final explicit content presented is equivalent, your correctness decision is unchanged.

Example Prompts (FRFP Style)

Reordering branches.

Prompt: “Swap the order in which you present the branches of the pipeline (symbolic vs numeric), but ensure that the final correctness decision I make remains invariant as long as the content is equivalent.”

This realizes the example in user-facing terms.

8 A.7 Conversation Pipeline Diagram (FRFP View)

Original Appendix Text

This final section informally illustrates how FRFP pipelines correspond to extended human–AI interaction.

Each exchange in a human–AI conversation can be seen as a full pipeline:

- **Representation Initiation (RI):** the human generates a new explicit query or instruction in E ;
- **Explicit Computation and Diagnostics (EC/ED):** the AI transforms the explicit state using explicit operations only;
- **Representational Backflow (RB):** the AI’s explicit output is presented to the human and enters tacit space T ;
- **Tacit Evaluation (TE):** the human interprets, evaluates, and grounds the result;
- **Human Final Decision (HFD):** the human decides whether to accept, reject, or request revisions, and may initiate a new explicit query.

Thus a conversation can be depicted schematically as:

$$RI \rightarrow (EC/ED)^* \xrightarrow{\text{AI explicit}} RB \rightarrow TE \rightarrow HFD \xrightarrow{\text{human tacit}} \text{new } RI \text{ in } E \rightarrow \dots$$

All AI activity remains inside the explicit category E ; all human interpretive activity remains inside the tacit category T . RB is the only crossing from E to T , and there is no morphism $T \rightarrow E$. New explicit instructions are modeled as fresh RI instances, not as tacit-to-explicit arrows.

Explanation (Plain Language)

A conversation is a sequence of FRFP pipelines:

1. You ask a question (RI).
2. The AI computes and diagnoses (EC/ED).
3. The result is shown to you (RB).
4. You interpret and evaluate (TE).
5. You decide how to proceed (HFD).
6. Your next message starts a new pipeline (fresh RI), based on your decision.

Damped Harmonic Oscillator Example

For DHO, a conversation might look like:

1. RI: “Here is my damped oscillator equation; please solve it.”
2. EC: AI derives a solution.
3. ED: AI checks the solution.
4. RB: AI presents solution and checks to you.
5. TE: you consider whether it matches your physical intuition.
6. HFD: you accept, reject, or request a revision.
7. If you request a revision, you issue a new RI which starts the next pipeline.

Example Prompts (FRFP Style)

Looping over conversation turns.

Prompt: “Based on your final solution and checks for my damped oscillator, I would like to revise the model to include a driving force. Start a new pipeline with the driven oscillator equation and repeat the same FRFP structure.”

Each such interaction is a new FRFP pipeline in the conversation sequence.

Tracking Notes

We now see the entire FRFP machinery instantiated as turn-based human–AI collaboration.